

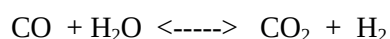
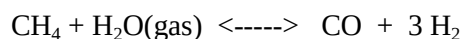
# 1,4-Butanediol Development Report

## Supporting Document for Assignment #3

The process development efforts to understand the reactions associated with producing 1,4-Butanediol from Propylene Oxide, Methane, Water and Oxygen have resulted in the following stoichiometry for each reaction:

For SynGas production, the reaction between water and natural gas, over the temperature and pressure ranges studied, has been determined to be well described by chemical equilibrium. Temperatures ranging from 750°C to 950°C and pressures from 1 to 2 MPa were studied. Note that CO<sub>2</sub> is formed simultaneously with the CO and H<sub>2</sub>. A Gibbs reactor may be used to describe the chemical equilibrium reactions in ASPEN Plus.

Reactions Occurring are as follows:



### Isomerization to form Allyl Alcohol:

Propylene Oxide -----> 0.83016 Allyl Alcohol + 0.049221 Acetone + 0.120615 Propionaldehyde  
(83.4% Conv. Propylene Oxide)

### Hydroformylation of Allyl Alcohol:

Allyl Alcohol + H<sub>2</sub> + 0.9658 CO -----> 0.8453 4-Hydroxy-1-Butyraldehyde (HBA) +  
0.1205 1-Hydroxy-2-Methylpropylaldehyde (HMP) +  
0.0342 n-Propyl alcohol (100% Conv Allyl Alcohol)

H<sub>2</sub>:CO:Allyl Alcohol Molar Feed Ratio: 6:2:1

### Hydrogenation:

4-Hydroxy-1-Butyraldehyde + H<sub>2</sub> -----> 1,4-Butanediol (100% Conv HBA)

1-Hydroxy-2-Methylpropylaldehyde + H<sub>2</sub> -----> 2-Methyl-1,3-Propanediol (MPD) (100% Conv HMP)

H<sub>2</sub>:Organic Molar Feed Ratio: 3:1

Reaction Temperatures and Pressures should be taken from the patent literature. If no conditions can be found, consult the instructor.